

Medical Image Computing

Summer of Code — Short Syllabus

Mentor's handbook (v1)

What this document is

A one-page-per-week checklist of topics, slide-deck pointers, and curated links. Use it *alongside* the longer `handbook_main.pdf`: read the relevant handbook chapter first for the big picture, then tick items off here as you work through them.

The full per-week plans (with exercises and notebooks) live in the repo at `week_N/Readme.md`.

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Week 1 — Image basics + probability & math foundations

Slide deck: `Slides_MIC_Introduction.pdf`

Topics. Pixel/voxel, intensity, colormap, signal vs. image, convolution and filtering, 2D Fourier basics; linear-algebra refresher (vectors, matrices, ranks, eigenvectors, SVD); norms $\ell^1, \ell^2, \ell^\infty$; probability (PDF/CDF, expectation, variance, Gaussian, Poisson, uniform); Bayes' theorem; MLE vs. MAP; gradient descent (intuitive).

Resources.

- 3Blue1Brown — *Essence of Linear Algebra*
- StatQuest — Probability vs Likelihood; Bayes' Theorem
- 3Blue1Brown — Convolution; Fourier Transform
- Strang — MIT OCW 18.06 lectures 1–5
- Tsitsiklis — MIT OCW 6.041 lectures 1, 2, 6, 7
- `nibabel` tutorial (medical image I/O)

Week 2 — Statistical priors (MRF/Gibbs) + image denoising

Slide decks: [Slides_MIC_ImageModeling.pdf](#), [Slides_MIC_ImageDenoising.pdf](#)

Topics. Random fields, neighbourhood systems, cliques; Markov Random Fields; Gibbs distribution & partition function; Hammersley–Clifford; Ising/Potts; continuous-valued GMRF; edge-preserving priors (Huber, Mumford–Shah); noise models in medical imaging (Gaussian, Poisson, Compound-Poisson, Rician, Rayleigh, K); SNR; Bayesian denoising = MAP estimation; non-local means; TV.

Resources.

- Stanford CS228 — [MRF notes](#)
- Geman & Geman (1984) — *Stochastic Relaxation, Gibbs Distributions, ...*
- Bouman — *Model-Based Image Processing* (free PDF, ch. 1–2, 7)
- Gudbjartsson & Patz — [Rician distribution in MRI](#)
- Buades et al. — [Non-local means](#)
- Steve Brunton — [Sparsity & \$\ell^1\$](#)

Week 3 — Inverse problems + CT and MRI physics

Slide decks: [Slides_MIC_ImageReconstruction.pdf](#), [Slides_MIC_XrayCT.pdf](#), [Slides_MIC_MRI.pdf](#)

Topics. Forward model $y = Gx + n$; ill-posedness; MAP reconstruction; Tikhonov (ℓ^2) and TV (ℓ^1) regularisation; matrix calculus (Jacobian, gradient of $\frac{1}{2}\|Gx - y\|^2$); complex derivatives & Cauchy–Riemann (MRI); gradient descent & line search; Beer–Lambert; attenuation coefficient & Hounsfield units; Radon transform; Fourier slice theorem; filtered back-projection; ART; Larmor precession; Bloch equations; T1/T2; k -space; compressed-sensing MRI.

Resources.

- Kak & Slaney — *Principles of Computerized Tomographic Imaging* (free PDF)
- Bouman — MBIP ch. 7–8 (link above)
- Computerphile — [How CT scans work](#)
- Allen Q. Ye — *MRI From Picture to Proton*
- iMaio — [k-space explainer](#)
- Lustig, Donoho & Pauly — [Sparse MRI \(CS-MRI\)](#)
- Petersen & Pedersen — [Matrix Cookbook](#)

Week 4 — Shape analysis + classical segmentation (+ registration, kernels)

Slide decks: [Slides_MIC_ImageSegmentation.pdf](#), [Slides_MIC_ShapeAnalysis.pdf](#), [Slides_MIC_Registration.pdf](#), [Slides_MIC_KernelMethods.pdf](#)

Topics. K-means (objective, alternating optimisation, NP-hardness, k-means++, k-medoids, Voronoi); Euclidean vs. Manhattan distances; GMM + EM; fuzzy C-means; spectral clustering (graph Laplacian); graph cuts; level sets (Chan–Vese); U-Net teaser; shapes as equivalence classes under similarity transforms; Procrustes + Generalised Procrustes; tangent-space PCA; Active Shape Models; Active Appearance Models; registration models (rigid / affine / B-spline / diffeomorphic) and similarity measures (SSD / NCC / MI); SVM & KKT; kernel trick; RKHS; Moore–Aronszajn; kernel PCA; Gram matrix.

Resources.

- StatQuest — [K-means; GMM/EM](#)
- von Luxburg — [Spectral clustering tutorial](#)
- Boykov & Funka-Lea — [Graph cuts in vision](#)

- Cootes et al. — [Active Shape Models](#)
- Pluim et al. — [MI registration survey](#)
- Boyd & Vandenberghe — [Convex Optimization](#) (KKT, ch. 5)
- Schölkopf, Smola & Müller — Kernel PCA paper

Week 5 — Convolutional neural networks for medical images

Slide deck: [Slides_MIC_DeepGenerativeModeling.pdf](#) (slides $\approx$ 1–30)

Topics. Convolution / cross-correlation, channels, stride, padding, dilation, receptive field; activations (ReLU, GELU, sigmoid, softmax, softplus); pooling & strided convolutions; normalisations (Batch / Group / Instance); losses (cross-entropy, focal, Dice, generalised Dice, Tversky); backprop & autodiff in one paragraph; SGD / Adam / AdamW & schedules; architectures (LeNet, AlexNet, VGG, ResNet, U-Net, nnU-Net, ViT, MedSAM); evaluation metrics (Dice, IoU, Hausdorff 95, ASSD, AUROC).

Resources.

- 3Blue1Brown — [What is a neural network? & backprop](#)
- Stanford CS231n — [class notes](#)
- PyTorch — [60-min blitz](#)
- Ronneberger et al. — [U-Net](#)
- Isensee et al. — [nnU-Net](#)
- MONAI — [tutorials](#)
- Maier-Hein et al. — [Metrics Reloaded](#)

Week 6 — VAEs and GANs (deep generative modelling)

Slide deck: [Slides_MIC_DeepGenerativeModeling.pdf](#) (slides $\approx$ 30–80)

Topics. Latent-variable models; ELBO; reconstruction & KL terms; reparameterisation trick; closed-form KL between Gaussians; β -VAE; VAE for anomaly detection; GAN min-max objective; Jensen–Shannon view; mode collapse, vanishing gradients; stabilisations (WGAN-GP, spectral norm, TTUR); conditional GANs (pix2pix, CycleGAN); medical applications (cross-modality synthesis, super-resolution); diffusion models (DDPM) at a high level; generative priors for Bayesian reconstruction (plug-and-play / score-based).

Resources.

- Kingma & Welling — [VAE](#)
- Lilian Weng — [From AE to \$\beta\$ -VAE](#)
- Goodfellow et al. — [GAN](#)
- Lilian Weng — [From GAN to WGAN](#)
- Gulrajani et al. — [WGAN-GP](#)
- Ho, Jain & Abbeel — [DDPM](#)
- Chung & Ye — [Score-based diffusion for MRI](#)

Week 7 — Paper reading + guided implementation

Slide deck: [Slides_ProjectTopicExamples.pdf](#)

Topics. How to read a paper (3-pass method); choosing a paper (venue / code / data / scope); one-page summary template; reproducing one experiment; baselines.

Starter paper pool (pick one).

- Aggarwal, Mani & Jacob — [MoDL \(model-based DL for inverse problems\)](#)
- Isensee et al. — [nnU-Net](#)

- Balakrishnan et al. — [VoxelMorph \(registration\)](#)
- Bhalodia et al. — [DeepSSM \(shape\)](#)
- Chung & Ye — [Score-based MRI](#)
- Ma et al. — [MedSAM](#)

Resources.

- Andrew Ng — [How to Read a Paper](#)
- Keshav — [3-pass paper-reading method](#)
- [Papers with Code — Medical](#)
- [Grand-Challenge](#) datasets
- [MedMNIST v2](#)

Week 8 — Final project

Deliverables. 6–10-page report; 10–15 slides; clean reproducible code with `requirements.txt` and run instructions; results/ folder with figures and tables; short live demo.

Suggested project shapes.

- Bayesian denoising with a learned prior (CNN denoiser vs. MRF MAP).
- Under-sampled MRI reconstruction (model-based / unrolled).
- Brain-tumour segmentation (BraTS; U-Net vs. nnU-Net).
- Shape-prior segmentation (ASM + CNN initialisation).
- Cross-modality synthesis (CycleGAN MR $\leftrightarrow$ CT).
- VAE anomaly detection on chest X-rays.

Common pitfalls. No baseline; no held-out test set; tiny test set without confidence intervals; no qualitative failure analysis; unreproducible code.

Resources.

- [slides/Slides_ProjectTopicExamples.pdf](#)
- [MONAI docs](#) and [tutorials](#)
- [Stanford CS230 final-project tips](#)
- [Distill — Research Debt](#)